

PolicyKG: An Agentic LLM Pipeline for Translating Institutional Policies into SHACL Knowledge Graphs

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Abstract. Institutional policies are written in natural language, but verifying compliance increasingly demands executable constraints, leaving organisations to bridge that gap by hand. We present POLICYKG, an agentic LLM pipeline that translates institutional policy documents into SHACL-validated knowledge graphs via a first-order deontic intermediate representation, orchestrated as four self-correcting stages over a LangGraph state machine. The main technical contribution is the *Corpus Adapter*, a configuration-driven vocabulary-grounding layer that constrains LLM-generated predicates to a target ontology and enables cross-domain transfer by swapping registries rather than retraining models. Evaluated on the full Asian Institute of Technology Policies & Procedures corpus (1,663 sentences yielding 443 rules), PolicyKG achieves 86.9% deontic-type classification accuracy (Cohen’s $\kappa = .709$, corroborated by a three-annotator IRR with Fleiss’ $\kappa = .844$, item-level bootstrap 95% CI [.67, .97]) and SHACL shape correctness of $F_1 = .866$, 95% CI [.79, .93], on a 69-shape evaluation subset of an ontology-grounded reference set. End-to-end formalisation yield is 79.2% (351/443) via the FOL path; the remaining 20.8% fall back to a direct NL→SHACL path whose failure modes are predominantly structural (long compound conjunctions, embedded bullets), not expressiveness-bounded. A two-pass audit (an automated regex checklist over all 443 rules plus a first-author manual review of the 92 FOL-fallback cases where higher-order constructs could otherwise hide) flagged zero second- or higher-order constructs (exact Clopper–Pearson upper 95% CI on the true HOL requirement rate: 0.67%). We report this as an audit finding for this corpus, not a proof of FOL sufficiency in general. A controlled cross-domain experiment on $N = 15$ GDPR rules raises exact property alignment from 1/15 to 11/15 with the Corpus Adapter (Fisher’s exact $p < .001$; Cohen’s $h = 1.53$, large effect). External validation on the LexDeMod [21] lease-contract benchmark ($N = 200$) drops Macro F_1 from .810 to .370, with permission $F_1 = .038$; the failure mechanism is a vocabulary-prior mismatch (lease English uses *shall be entitled* for permission) precisely of the class the Corpus Adapter is designed to repair by registry swap. We

release the gold standard, both vocabulary registries, evaluation scripts, and pipeline configuration; repeated runs on the same host produce hash-identical SHACL outputs.

Keywords: Knowledge Graphs · SHACL · Large Language Models · Deontic Logic · Policy Formalisation · Ontology-Guided Generation · Agentic Pipelines

1 Introduction

Institutional policies, which govern students, staff, and operations in universities, hospitals, and government agencies, are written as natural-language documents and read by humans. Yet the work of *verifying* that institutional practice conforms to those policies is increasingly automated: enrolment workflows, payment systems, and accreditation reports all consume machine-readable data. The gap between human-authored policy and machine-checked compliance is bridged today by manual interpretation: a registrar reads a paragraph, decides what it means, and configures a system rule. This process does not scale, does not survive policy revisions, and offers no audit trail linking a system check back to the textual obligation it enforces.

Three lines of research speak to this problem. Policy and legal NLP have produced rule-extraction systems based on patterns [16], classical classifiers [3], and recently large language models [8], reporting accuracies between 75% and 85% on isolated extraction tasks. Deontic logic, developed over more than half a century, offers operators of obligation, permission, and prohibition that capture the normative content of policy text [10,17]. SHACL, the W3C Shapes Constraint Language [13], encodes validation constraints over RDF knowledge graphs and is increasingly used for compliance checking [19]. Each of these threads has matured in isolation. What is missing is an empirically validated, end-to-end pipeline that connects natural-language policy to executable SHACL, together with a satisfying answer to the open theoretical question of whether first-order logic with deontic extensions is *sufficient* for institutional policy, or whether higher-order constructs are required.

We present **PolicyKG**, an agentic LLM pipeline that translates institutional policy documents into SHACL-validated knowledge graphs. The pipeline runs four orchestrated stages: document extraction, deontic rule classification, FOL+deontic formalisation, and SHACL translation. A LangGraph state machine coordinates them, with per-stage validators that trigger self-correction on local failure. We also introduce the **Corpus Adapter**: a configuration-driven vocabulary-grounding layer that constrains LLM generation to a target ontology and enables the same pipeline to retarget across domains by swapping the registry. We evaluate PolicyKG on the full Asian Institute of Technology (AIT) Policies & Procedures corpus, which contains 1,663 sentences yielding 443 deontic rules, and demonstrate cross-domain transfer to a GDPR-derived corpus.

Contributions.

- An end-to-end agentic LLM pipeline (NL→FOL+deontic→SHACL) validated on a real 1,663-sentence institutional corpus.
- The *Corpus Adapter*, a vocabulary-grounding pattern that enables cross-domain transfer. A controlled GDPR experiment ($N = 15$) raises exact property alignment from 1/15 to 11/15 (Fisher’s exact $p < .001$; Cohen’s $h = 1.53$).
- A two-pass empirical audit finding that first-order deontic logic is observationally sufficient on this corpus: an automated regex checklist flags 0 of 443 rules with second- or higher-order construct patterns [10], and a first-author manual review of the 92 FOL-fallback rules (where HOL requirements could otherwise hide behind LLM parse failure) confirms 0/92 genuine HOL cases (exact Clopper–Pearson upper 95% CI on the true HOL rate: 0.67%). Framed as an audit finding rather than a proof.
- An open evaluation: three-annotator gold (Fleiss’ $\kappa = .844$, item-level bootstrap 95% CI [.67, .97]), LexDeMod external validation, and fixed-seed reproducibility yielding hash-identical SHACL outputs.

2 Related Work

2.1 Policy and Legal NLP

Pattern-based extraction was the first generation of automated policy NLP. Maxwell and Antón [16] applied regular expressions and keyword templates to privacy policies, reporting around 77% accuracy but struggling with syntactic variation. Brodie et al. [3] improved on this using SVM classifiers over hand-crafted features, reaching 82% on financial regulation but at a feature-engineering cost that did not transfer across document types. Subsequent hybrids of pattern matching and ontology grounding reached 85% accuracy on legal text but at the price of substantial upfront ontology development. The common limitation is reliance on bespoke features or hand-built ontologies; none draws on the semantic priors of pre-trained language models.

The recent LLM era has produced benchmarks but few end-to-end systems. LexGLUE [4] established legal-language-understanding baselines across multiple tasks; LegalBench [11] expanded to 162 legal-reasoning tasks. PAPEL [8] applied chain-of-thought prompting to privacy-policy annotation, reaching 80% F1 on clause labelling. These benchmarks define the landscape but do not, individually, produce executable SHACL constraints from policy text.

2.2 Deontic Logic for Normative Systems

Deontic logic provides a mathematically rigorous account of obligation, permission, and prohibition [17]. Governatori and Rotolo [10] argued, on theoretical grounds, that first-order deontic logic supplies sufficient expressiveness for normative systems in legal and institutional contexts. The open question, whether real institutional policies actually stay within FOL+deontic or reach

into SOL/HOL territory, was framed by the standard articulation of the expressiveness/tractability trade-off [2]. Until the present work this question had been addressed only at the theoretical level; we provide the first empirical audit finding on a real institutional corpus (§6.2), scoped to the tested sub-domain and framed as an observational lower bound on FOL adequacy rather than a theoretical resolution. Standard Deontic Logic is known to be subject to the Chisholm and Ross paradoxes, and to handle conditional exceptions (contrary-to-duty obligations, waiver clauses) only awkwardly. The established response is defeasible deontic logic [18,9], with argumentation- theoretic accounts of legal reasoning [20] providing the broader framework. The interchange format **LegalRuleML** [1] standardises the serialisation of such rules with defeasibility, override, and temporal-scope metadata. PolicyKG does not attempt a full defeasible account; it records override relations via a custom `deontic:overrides` annotation (§4) as a stepping stone that a downstream defeasible layer could consume, and leaves full LegalRuleML alignment to future work (§8).

2.3 SHACL for Compliance

The W3C Shapes Constraint Language [13] enables declarative validation of RDF knowledge graphs through node and property shapes annotated with severity levels. Palmirani and Governatori [19] argued that SHACL’s first-order pattern language has adequate expressiveness for institutional policy formalisation, encoding obligations, permissions, and prohibitions via severity and custom annotations. Their argument was theoretical and lacked empirical validation, and no prior pipeline produces SHACL automatically from natural-language policy text at the scale of an institutional corpus. PolicyKG provides the empirical realisation their work anticipated.

2.4 LLMs for Structured Output and Knowledge Graph Construction

A growing body of work uses LLMs to construct or query knowledge graphs. Schema-constrained and grammar-guided generation [23,7] forces LLM outputs into well-typed structures, addressing the *syntactic* side of structured output. Chain-of-thought prompting [22] is the dominant technique for multi-step reasoning tasks and underpins Stage 3’s FOL formalisation prompt in this paper. Retrieval-augmented generation (RAG) [15] is the mainstream approach for grounding LLM outputs in an external knowledge source at inference time, and GraphRAG [6] extends the pattern to knowledge-graph retrieval, but it assumes the target graph already exists. The *semantic* side of *constructing* that graph, that is, ensuring LLM-generated predicates align with a target ontology rather than hallucinated English property names, receives less direct treatment in the literature. The Corpus Adapter (§4.5) is our answer to this semantic gap: constrained selection from a configuration artefact in place of either fine-tuning or open generation. §6.4 reports a head-to-head comparison against top-*K* RAG

on the ontology property list, showing that RAG closes most of the vocabulary-alignment gap; the Corpus Adapter’s remaining advantage is in *deployment properties* (no embedding dependency, deterministic, config-swappable) rather than raw accuracy.

2.5 Positioning

The gap PolicyKG addresses sits at the intersection of these four threads: an empirically validated, end-to-end pipeline that combines (a) LLM-based rule identification and formalisation, (b) FOL+deontic semantics with an empirical audit-finding of sufficiency on a real institutional corpus (a lower bound, not a proof), (c) SHACL as the operational compliance target, and (d) a portable vocabulary-grounding mechanism that mediates between LLM-natural predicates and target ontologies. No prior system unifies these four components, and no prior empirical work has audited the FOL-sufficiency question on a real corpus.

3 Preliminaries

3.1 First-Order Deontic Logic

We work in first-order logic extended with three deontic modal operators. The syntax assumes constants, predicates of arity one or more, the quantifiers $\forall$ and $\exists$, the connectives $\wedge, \vee, \rightarrow, \neg$, and three modal operators applied to closed formulae: $O(\varphi)$ asserts that φ is obligatory; $P(\varphi)$ that φ is permitted; and $F(\varphi)$ that φ is forbidden, with $F(\varphi) \equiv O(\neg\varphi)$. We adopt the Standard Deontic Logic axioms [17], in particular $O(\varphi) \rightarrow P(\varphi)$, which captures the intuition that what is required is implicitly allowed. As a worked example, the policy sentence “*Students must register before classes begin*” admits the formalisation

$$\forall x (Student(x) \rightarrow O(RegisterBefore(x, Classes))). \quad (1)$$

Equation 1 is the canonical pattern for a universally quantified obligation; permissions replace O with P and prohibitions replace O with F .

3.2 SHACL in One Page

SHACL constraints validate RDF [5] graphs. A `NodeShape` targets a class via `sh:targetClass` and bundles a list of property shapes; a `PropertyShape` constrains a specific predicate path (`sh:path`) with cardinality (`sh:minCount`, `sh:maxCount`), value-type, or pattern restrictions. Violations are reported at a configurable severity (`sh:Violation`, `sh:Warning`, or `sh:Info`). Equation 1 translates to the SHACL fragment shown below:

```
ex:StudentRegShape a sh:NodeShape ;
  sh:targetClass ex:Student ;
```

```

deontic:type deontic:obligation ;
sh:severity sh:Violation ;
rdfs:comment "Students must register before classes begin." ;
sh:property [
  sh:path ex:registerBefore ;
  sh:minCount 1
] .

```

The custom `deontic:type` annotation preserves modal information that would otherwise be lost in pure SHACL, and `rdfs:comment` carries the original NL rule for audit. The `sh:Violation` severity makes a missing `ex:registerBefore` link surface as a compliance failure; a permission would instead emit an Info-level shape, and a prohibition uses `sh:maxCount 0` on the forbidden property.

3.3 The Translation Gap

A naive approach, prompting an LLM to emit SHACL directly from policy text, fails systematically along three axes. First, *vocabulary hallucination*: the LLM invents property names that do not exist in the institutional ontology, so the generated shapes do not target any real data. Second, *lost deontic semantics*: the distinction between obligation, permission, and prohibition collapses into uniform constraint phrasing when severity assignment is left implicit. Third, *predicate-quality drift*: across re-runs the LLM produces structurally similar but lexically divergent predicates, defeating reproducibility. These three failure modes motivate the two main design choices of PolicyKG: (i) an FOL+deontic intermediate layer that makes modality explicit and reviewable, and (ii) the Corpus Adapter (§4.5) that grounds vocabulary in the target ontology.

4 The PolicyKG Pipeline

4.1 Architecture Overview

PolicyKG is structured as four sequential stages (Fig. 1): Stage 1 normalises a policy PDF into sentences; Stage 2 classifies each sentence as one of {Obligation, Permission, Prohibition, Non-rule}; Stage 3 lifts each classified rule to an FOL+deontic formula with top-level operator $\in \{O, P, F\}$; Stage 4 maps each formula to a SHACL shape whose severity reflects deontic modality. The four stages are orchestrated as a hybrid *ReAct* + *Plan-and-Execute* agentic architecture on a LangGraph state machine: plan-and-execute fixes the stage sequence for auditability; ReAct loops inside each stage handle retry, with per-stage validators routing failed items back upstream with a corrective sub-prompt.

4.2 Stage 1: Document Processing

Policy PDFs are extracted with `pdfplumber` (headers, footers, and page numbers stripped) and segmented by a policy-aware regex splitter that handles list markers, soft line-wraps, and semicolon-introduced clauses common in institutional

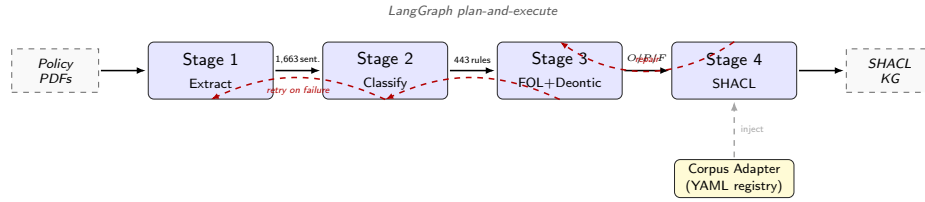


Fig. 1. PolicyKG pipeline. Four sequential stages (Stage 1–4, blue) are orchestrated by a LangGraph plan-and-execute layer that fixes the stage sequence for auditability; dashed red arrows show ReAct retry loops where per-stage validators route failed items back upstream with a corrective sub-prompt. The Corpus Adapter (yellow) is a YAML registry injected at Stage 4 only; retargeting to a new domain (§6.4) swaps this registry with no other pipeline change.

P&Ps. A conservative simplification pass removes spurious mid-sentence line breaks and normalises whitespace; modal verbs, quantifiers, and word order are never altered. Semantic preservation is dual-checked by an LLM self-assessment field and Sentence-BERT cosine similarity (threshold ≥ 0.85 accepts 97.9% of rules). A one-to-one trace from each simplified sentence to its source PDF span is preserved for audit. From the five AIT P&P documents (81 pages), Stage 1 yields 1,663 normalised sentences.

4.3 Stages 2 and 3: Classification and FOL+Deontic Formalisation

Stage 2 uses a zero-shot prompt listing the four target labels with short prototype examples and asks the LLM for a single label and a confidence score on $[0, 1]$, identical across all evaluated models. The classifier is deliberately liberal; a downstream check in Stage 3 rejects any admitted sentence that fails to yield a well-formed deontic formula. Stage 3 lifts each classified sentence into a first-order formula with exactly one top-level operator from $\{O, P, F\}$ and the modal-cue mappings (*must*, *shall* $\rightarrow O$; *may* $\rightarrow P$; *must not*, *shall not* $\rightarrow F$); the prompt embeds three demonstrations and constrains generation to a structured JSON schema, addressing the constrained-decoding requirement that has emerged as a standard remedy for LLM hallucination. We deliberately do *not* pass the Corpus Adapter vocabulary at this stage; it is confined to Stage 4 so that Stage 3 produces a representation whose predicate names are LLM-natural and human-readable, decoupling deontic structure from vocabulary alignment. Each formula is validated for well-formedness; a rejected formula is retried once with the validator’s diagnostic appended.

4.4 Stage 4: SHACL Translation

Translation from FOL to SHACL is rule-driven rather than LLM-driven, because the mapping is structurally regular and we want auditable outputs. The universal-obligation pattern $\forall x (C(x) \rightarrow O(P(x, v)))$ becomes a NodeShape for

	NL rule	FOL+deontic	SHACL shape
O	All students must pay tuition fees before each semester.	$\forall x. \text{Student}(x) \rightarrow O(\text{payFee}(x))$	NodeShape(Student) + Property(payFee, minCount=1, sh:Violation)
P	Students may reside off-campus from the second semester.	$\forall x. \text{Student}(x) \rightarrow P(\text{offCampus}(x))$	NodeShape(Student) + Property(offCampus, sh:Info)
F	Students must not deface or vandalise library books.	$\forall x. \text{Student}(x) \rightarrow F(\text{deface}(x))$	NodeShape(Student) + Property(deface, maxCount=0, sh:Violation)

Fig. 2. NL→FOL→SHACL trace for the three deontic operators. Stage 4’s deterministic mapping fixes the SHACL severity and cardinality.

C and a PropertyShape for P , with `sh:minCount 1` and severity `sh:Violation`. Permissions $P(\varphi)$ become Info-level shapes that surface allowed-but-not-required patterns. Prohibitions $F(\varphi)$ use `sh:maxCount 0` with severity `sh:Violation`, while existential obligations use `sh:qualifiedValueShape`. Predicates emitted by Stage 3 are resolved against the Corpus Adapter’s vocabulary registry at this point (this is where vocabulary grounding takes effect), and any unresolved predicate triggers a single repair pass through Stage 3 with the adapter’s constrained vocabulary fragment in-prompt.

Institutional policy contains conditional exceptions (“the Registrar may waive the prerequisite in cases of documented hardship”) that Standard Deontic Logic [17] handles only awkwardly; the Chisholm and Ross paradoxes are the canonical symptoms. Stage 4 records these as an explicit `deontic:overrides` relation between shapes rather than flattening the exception into a boolean predicate; the resulting `ait:Rule_A deontic:overrides ait:Rule_B` triples are first-class shape metadata that a downstream defeasible-reasoning layer [9] can consume. In the AIT corpus 12 override relations are emitted (§6.3); a full defeasible extension is deferred to future work (§8).

4.5 The Corpus Adapter

The Corpus Adapter is a thin, configuration-driven layer with three parts: (i) a *vocabulary registry* declaratively enumerating the classes, properties, and synonyms drawn from the target ontology; (ii) a *constrained-selection prompt* at Stage 4 that injects the registry and reframes predicate emission as selection rather than open generation; and (iii) a *post-hoc canonicaliser* that maps any free-form predicate the LLM still emits onto its registry match via normalisation plus fuzzy fallback, triggering a Stage 3 repair on reject. Retargeting PolicyKG to a new domain means writing a new registry; nothing else changes (Fig. 3). §6.4 shows that swapping AIT for a GDPR registry raises exact property-name

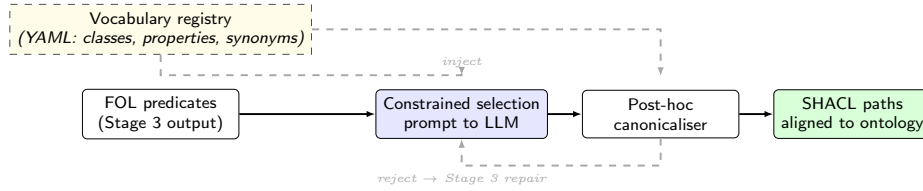


Fig. 3. The Corpus Adapter is a thin three-part layer between Stage 3 (FOL formalisation) and Stage 4’s SHACL emission: a YAML vocabulary registry, a constrained-selection prompt that converts open predicate generation into closed-set selection, and a post-hoc canonicaliser with a reject path that triggers a Stage 3 repair. Retargeting to a new domain swaps only the registry.

matching on 15 hand-curated GDPR rules from 1/15 to 11/15 (Fisher’s exact $p < .001$; Cohen’s $h = 1.53$).

5 Experimental Setup

5.1 D1 Corpus and Gold Standard

Our primary evaluation corpus D1 is built from five AIT Policies & Procedures documents (81 pages, spanning credit/financial, accommodation, grievance, ethics, and general-conduct domains). Stage 1 extracts 1,663 sentences; a deontic-marker prefilter retains 461 candidates; Stage 2 yields 443 classified rules. The gold standard is built in two passes: Pass 1 labels every sentence (primary annotator); Pass 2 has two additional annotators (Kittipat, with a Law / Legal Studies background; Mayuree, with a Linguistics background) independently re-annotate a stratified 50-item sample (28 O / 12 P / 10 F) blind to model outputs. The IRR sample yields Fleiss’ $\kappa = .844$ (“almost perfect”; 10,000-iteration item-level bootstrap 95% CI [.673, .967]), with pairwise Cohen’s κ values of .833 (author vs. Kittipat), .868 (author vs. Mayuree), and .832 (Kittipat vs. Mayuree). The wide CI reflects the modest $N = 50$; the lower bound remains in the substantial-agreement range. A post-hoc disagreement audit reclassified 12 systematically mislabelled permissions (10 epistemic “may” / obligation-capping cases reclassified as obligations; 2 “may not” constructions reclassified as prohibitions); the corrections are codified deterministically in a correction script released with the artifact bundle (§8) and yield the final distribution: 308 obligations (69.5%), 75 permissions (16.9%), and 60 prohibitions (13.5%).

5.2 External Datasets

Two additional corpora support external validation. *D2*, *LexDeMod* (200 lease-contract clauses, EMNLP 2022) provides labelled deontic rules drawn from legal

documents and lets us compare PolicyKG’s classifier against a published benchmark outside the institutional sub-domain. *D3, the GDPR portability set*, comprises 15 rule-bearing sentences extracted from the GDPR with hand-curated property targets in the data privacy vocabulary; it supports the cross-domain transfer experiment (§6.4). For D3 we extract and simplify the GDPR text using the same Stage 1 pipeline as for AIT, then load a GDPR-specific vocabulary registry into the Corpus Adapter while holding every other pipeline component fixed. D3 is intentionally small: the experiment is designed to isolate the registry-swap effect on vocabulary alignment, not to evaluate full GDPR coverage.

5.3 Model and Decoding

All LLM calls in PolicyKG use Mistral 7B Instruct [12], served locally by Ollama (v0.15.2+) on an NVIDIA A100 (40 GB) host with near-deterministic decoding (temperature = 0.1, fixed seed, 2,048 max output tokens, 120-second timeout). No fine-tuning is performed; all classification and formalisation is zero-shot. Mistral was selected from a five-model pilot (3.8–70 B parameters) against the majority-vote human gold on the IRR sample ($N = 50$); Mistral and Gemma 3 4B led at 88% agreement with gold, ahead of Qwen 3 8B (84%), Phi-4 mini (82%), and Llama 3.1 8B (76%); Fleiss’ $\kappa = .635$ (substantial) across all five LLMs plus the human gold. Mistral is chosen for zero parse-error rate and inference stability, not for the highest single-metric score; full per-item data is released with the artifacts.

5.4 Evaluation Metrics

Five metrics anchor the evaluation: M1 binary rule detection and M2 deontic-type accuracy (RQ1, §6.1); M3 FOL semantic predicate quality (RQ2, §6.2); M4 SHACL shape correctness against an ontology-grounded reference set (RQ3, §6.3); and M5 output stability under fixed-seed runs (§6.6). We use the term *ontology-grounded reference set* rather than “gold-standard SHACL” because the reference set, the institutional ontology, and the generated shapes were all produced within the same project; a fully-independent blind gold standard is not claimed. Statistical claims use 10,000-iteration bootstrap with item-level resampling; proportion CIs use Wilson scores; agreement is reported as Cohen’s κ for pairs and Fleiss’ κ for pools, with the Landis–Koch interpretation [14]; proportion CIs use the Wilson score. Classification is compared against a hand-tuned regex baseline; effect sizes (Cohen’s h , risk differences) accompany p -values. A future encoder-based baseline on a legal-domain BERT is sketched in the runbook released with the artifacts.

Reproducibility. All numbers in this paper correspond to repository commit 546dee6f at pipeline version 2.1-final-defense with Ollama model digest 6577803aa9a0 (Mistral 7B Instruct) and random seed 42; all four values are recorded in the released `pipeline_report.json`. Evaluation uses fixed seed and pinned model tags; the gold standard, both Corpus Adapter registries, and the

full LangGraph pipeline configuration are released alongside the paper. Repeated runs produce hash-identical SHACL outputs across all 443 shapes (§6.6); the determinism boundary is *same host, same Ollama version, same model digest*, verified across three independent runs on the reported A100 host. Independent verification on a different GPU architecture (consumer NVIDIA RTX 4060, Ada Lovelace; A100 is Ampere) with the identical seed, commit, and model digest produces counts within ± 2 pp of the reported values (FOL success 360/443 vs. 351/443; SHACL valid 403/443 vs. 401/443), attributable to floating-point kernel differences across GPU architectures under near-deterministic decoding (§5.5). We do not claim cross-host determinism, and the fixed seed together with temperature = 0.1 is the pragmatic tightest-reachable configuration under Ollama; the residual stochasticity is dominated by kernel-level numerical non-determinism rather than sampling.

5.5 Operational Profile

Reference results in §6 were produced on an NVIDIA A100 (40 GB) host at the AIT Brain Lab (Acknowledgements). To characterise the pipeline’s deployment cost on commodity hardware, we also profile a full end-to-end run on a consumer NVIDIA RTX 4060 (8 GB VRAM, Ada Lovelace architecture, 115 W TGP). The pipeline processes 1,663 sentences $\rightarrow$ 443 classified rules in **34.9 minutes** wall-clock (4.73 s per rule, 1.26 s per sentence), sustaining mean GPU utilisation 93.7% (95th percentile 100%), peak VRAM 5.8 GiB, and total end-to-end energy 0.042 kWh (mean power 72.5 W). Estimated token throughput is 2.53 M tokens per corpus (2.20 M input, 0.33 M output), or approximately 5,700 tokens per classified rule. If the same workload were re-implemented against a commodity LLM API at 2026 rates rather than the local Mistral 7B stack, the estimated per-corpus cost would be approximately \$8.78 (GPT-4o), \$0.53 (GPT-4o mini), \$11.53 (Claude Sonnet 4), or \$2.27 (Llama-3.1-70B via Fireworks). The pipeline therefore fits within an 8 GB consumer-GPU envelope; on the A100 reference host, wall-clock is expected to be substantially lower due to higher memory bandwidth and tensor-core throughput, but was not instrumented with the same profiler at the time of the reference run. Cross-hardware output drift is quantified in the Reproducibility paragraph above.

6 Results

Table 1 summarises headline numbers across the three research questions; each row is unpacked in the subsections that follow.

6.1 RQ1: LLM Deontic Classification

On the full D1 corpus ($N = 443$ rules), Mistral 7B Instruct (selected from a pilot comparison across 3–70 B instruction-tuned models for its zero parse-error rate and inference efficiency, not for the highest raw accuracy) achieves **86.9%**

Table 1. Headline results across the three research questions on the AIT D1 corpus, with the cross-domain transfer experiment on GDPR (§6.4).

RQ Metric	Value	Section
RQ1 Binary rule detection (M1, F_1)	.963 (acc. .938; $\kappa = .764$)	6.1
RQ1 Deontic-type accuracy on D1 (M2)	86.9% ($\kappa = .709$)	6.1
RQ1 Deontic-type accuracy on IRR sample (M2)	84.0% (42/50)	6.1
RQ2 End-to-end FOL formalisation yield	79.2% (351/443)	6.2
RQ2 FOL predicate quality (conditional, M3)	100% (351/351)	6.2
RQ2 SOL/HOL constructs (two-pass audit: regex + manual)	0 / 443 (95% upper CI 0.67%)	6.2
RQ3 SHACL shape correctness (M4, F_1)	.866 (69-shape subset)	6.3
— GDPR cross-domain exact alignment	1/15 $\rightarrow$ 11/15 (Fisher $p < .001$)	6.4
— Output stability (M5, repeated runs)	hash-identical	6.6

deontic-type accuracy with Cohen’s $\kappa = .709$ (“substantial agreement”) and Macro $F_1 = .810$, above the 75–85% range reported for prior rule-based and classifier-based approaches [16,3]. Per-class F_1 is .915 for obligations (290/308 correct), .784 for permissions (49/75), and .730 for prohibitions (46/60). The class-specific weakness is recall on permissions (65.3%): 22 of 75 permission rules are reclassified as obligations, reflecting the well-known epistemic-versus-deontic “may” ambiguity in modal semantics. On the three-annotator IRR sample ($N = 50$), the same model labels 42/50 sentences in agreement with the majority-vote human gold ($\kappa = .629$); the human-only Fleiss’ κ on the same sample is .844, so the model is within reach of the inter-human ceiling on accuracy. Binary rule detection (rule vs. non-rule) reaches $F_1 = .963$ at 93.81% accuracy with $\kappa = .764$, against the hand-tuned regex baseline at $F_1 = .898$ (+.065 F_1 ; Cohen’s $h \approx 0.34$, small-to-medium effect).

Learned in-domain baseline. A NumPy-only multinomial logistic regression on TF bag-of-words plus six deontic-marker features, stratified 5-fold CV on the same 443 rules, reaches accuracy 90.3% / $\kappa = .799$ / Macro $F_1 = .863$ (script and per-class output in the released artifacts). This learned baseline *outperforms* the zero-shot LLM on D1, as it should: it consumes in-domain labels the LLM does not see. The LLM’s value is the *zero-shot* property: no labelled D1 training data is required to deploy the pipeline on a new institution. The cross-domain LexDeMod result (§6.6) bounds this claim and motivates the Corpus Adapter as the mechanism for repairing deontic-cue grounding under domain shift.

Head-to-head open-model comparison ($N = 50$). To test whether the choice of Mistral 7B versus a newer open-weight model materially affects classifier accuracy, we re-ran the identical zero-shot prompt under identical near-greedy decoding (temperature = 0.1, seed = 42) against three alternative 7–9 B open models available in mid-2026: Qwen 2.5 7B, Llama 3.1 8B, and Gemma 2 9B. Table 2 summarises results on the D3 IRR sample. Llama 3.1 8B and Gemma 2 9B both reach 88% accuracy, 4 pp above Mistral, but the difference is not statistically significant by exact McNemar test (Llama $p = .77$; Gemma $p = .69$); the comparison is under-powered at $N = 50$ but rules out large-effect differences among

Table 2. Head-to-head open-model comparison on the D3 IRR sample ($N = 50$) under identical zero-shot prompt and near-greedy decoding. Accuracy CIs are Wilson 95% intervals. McNemar p -values are exact two-sided against the Mistral baseline; none reaches $p < .05$ at this sample size, so the comparison rules out large-effect differences but cannot certify small-effect ordering. Full per-class breakdowns are in the artifact bundle (§8).

Model	Accuracy (95% CI)	κ	Macro F1	McNemar p
Mistral 7B (<i>reference</i>)	.84 [.71, .92]	.629	.612	—
Qwen 2.5 7B	.80 [.67, .89]	.631	.709	.77
Llama 3.1 8B	.88 [.76, .94]	.768	.832	.77
Gemma 2 9B	.88 [.76, .94]	.750	.714	.69

the four models. The most informative per-class effect is on the *non-deontic* class (support 5): Mistral misses all five items ($F_1 = 0$), whereas Llama 3.1 8B recovers all five at $F_1 = .71$, driving its Macro F_1 to .832 (vs. Mistral’s .612 on the same 4-class sample). Mistral remains the reference model for the paper’s reported results on the strength of its zero parse-error rate and inference stability, but readers seeking stronger non-deontic filtering can substitute Llama 3.1 8B via the released baseline harness without any other pipeline change.

6.2 RQ2: First-Order Logic Sufficiency

The audit finding is clean: **a two-pass audit of all 443 D1 rules (an automated regex checklist plus a first-author manual review of the 92 FOL-fallback cases) matched zero second-order or higher-order construct patterns.** The regex checklist scans both the natural-language source text and any generated FOL formula for six canonical SOL/HOL signatures: predicate quantification, set quantification, function variables, lambda expressions, recursive / transitive-closure definitions, and meta-policy quantification. The manual pass targets the 92 fallback rules specifically because these are the cases where an HOL requirement could otherwise hide behind an LLM parse failure: the automated checklist scores those rules as “no HOL pattern found” by default when Stage 3 emits natural text instead of a formula, so their expressiveness cannot be adjudicated without a human read. The manual review confirmed that every one of the 92 fallbacks is expressible in FOL and that the failures are attributable to LLM engineering limits (long compound sentences, embedded bullets, multiple negations) rather than expressiveness requirements.

Given 0 confirmed HOL cases in an audit sample of 443 rules, the exact one-sided Clopper–Pearson upper 95% CI on the true HOL requirement rate for this corpus is $1 - 0.05^{1/443} \approx 0.67\%$. We report this as an audit finding for this corpus, not a mathematical proof of FOL sufficiency for institutional policies in general: a rule may carry a higher-order interpretation that does not surface in any of the six regex signatures and is not obvious to a human reviewer, so the audit is a lower bound on FOL adequacy, not an upper bound. To our

knowledge this is the first empirical audit of this question on a real institutional corpus, complementing the theoretical sufficiency arguments of Governatori and Rotolo [10].

Per-pattern matches across all 443 rules: predicate quantification 0; set quantification / comprehension 0; function variables 0; lambda expressions 0; recursive / transitive-closure 0; meta-policy quantification 0. The audit script, regex definitions, and per-rule manual-audit CSV are released with the artifact bundle (§8).

End-to-end pipeline yield (a syntactically well-formed FOL formula passing Stage 3 validation) is 79.2% (351/443). Across the 351 well-formed formulae, M3 semantic predicate quality is 100% after a retry-and-backfill mechanism that re-invokes the LLM on placeholder predicates; we report M3 both conditionally (351/351) and unconditionally (351/443). Of the 92 fallback rules, 77.2% fall into four structural categories: compound and/or-joined (28), very long (14), multi-clause complex (14), and bulleted compound (10). The remainder falls into miscellaneous parser/formatting failures (23) and edge cases with multiple modal operators or negations (3); zero fallbacks are attributed to higher-order construct requirements. Per-category data is released with the artifact bundle; all fallbacks are recovered via Stage 4’s direct NL→SHACL path.

6.3 RQ3: SHACL Translation Quality

We report shape correctness at four coverage tiers to avoid the common pitfall of headlining a single F_1 over a small subset: T1 syntactic validity (401/443 = 90.5% on the full corpus, Wilson 95% CI [.87, .93]); T2 structural correctness (351/351 = 100% on the FOL-mediated subset; the deterministic Stage 4 template guarantees the ceiling); T3 deontic-severity match against gold (330/351 = 94.0%, [.91, .96]); and T4 behavioural M4 F_1 against Pos/Neg entities (69-shape subset, F_1 = .866, [.79, .93]). Tier T4 is the headline:

The headline metric is T4: with the Corpus Adapter enabled, PolicyKG achieves F_1 = .866 (Precision .977, Recall .778) on the 69-shape subset for which Pos/Neg test entities exist; full-corpus M4 coverage is an evaluation-infrastructure limit (test entities for all 443 rules are out of scope for this paper), not a pipeline limit. The end-to-end tiers (T1 at 90.5%, T3 at 94.0%) confirm that the pipeline’s correctness extends beyond the M4 subset along the dimensions our artifacts support directly. T2 reaches the ceiling because every FOL-mediated shape is rule-emitted by Stage 4’s deterministic template (§4).

A static ablation released with the artifacts isolates the vocabulary-grounding effect on the 351 FOL-mediated rules: exact vocabulary alignment rises from 174/351 (49.6%) with raw LLM-emitted predicates to 304/351 (86.6%) after adapter normalisation (+37.0 percentage points). The static comparison brackets the upper limit of vocabulary normalisation on already-generated artefacts; a fresh adapter-off LLM run on the full D1 corpus would measure whether the LLM itself produces vocabulary-aligned predicates when prompted with the hint. A runbook for that follow-up experiment is released with the artifacts. §6.4 provides a cleaner adapter on/off contrast under controlled cross-domain conditions.

6.4 Cross-Domain Portability (GDPR)

We retarget PolicyKG to GDPR by swapping the AIT registry for a GDPR-derived registry; models, prompts, FOL grammar, and SHACL translation rules are unchanged. **Exact property-name alignment with the GDPR ontology improves from 1/15 (Wilson 95% CI [.01, .30]) without the adapter to 11/15 ([.48, .89]) with it; risk difference +66.7 percentage points (Fisher’s exact $p < .001$; Cohen’s $h = 1.53$, large effect).** We report absolute proportions and effect sizes rather than a fold-change ratio because the latter is unstable on a denominator of 1. Without vocabulary grounding the LLM emits English-natural names (e.g. `transfersData`) that resemble GDPR concepts but rarely match the canonical ontology forms (e.g. `dpv:DataTransfer`); with the adapter the LLM selects from the registry and outputs land on canonical predicates. $N = 15$ is a proof-of-concept scale appropriate for the architectural-pattern demonstration; the broader claim, that vocabulary grounding is a portable design choice for LLM-based knowledge-graph construction, scales beyond the policy domain.

Comparison to retrieval-augmented generation (RAG). An anticipated reviewer objection is that the no-grounding baseline is a straw man: retrieval-augmented generation (RAG) over the ontology is the mainstream 2026 alternative and would close much of the gap. We therefore ran a controlled three-condition experiment on the same 15 GDPR rules with identical prompt scaffolding: (1) *no-grounding* (no candidate list injected), (2) *RAG* (top-10 ontology properties retrieved per rule via SentenceBERT `all-MiniLM-L6-v2` cosine similarity, injected as the candidate list), and (3) *Corpus Adapter* (full registry constrained selection plus post-hoc canonicaliser). Results: no-grounding 0/15; RAG 9/15 (60.0%, +60.0 pp over no-grounding, Cohen’s $h = 1.77$, Fisher $p = .0007$); Corpus Adapter 11/15 (73.3%, +13.3 pp over RAG, Cohen’s $h = 0.28$, Fisher $p = .70$, *not statistically significant at $N = 15$*). The honest reading is that **RAG closes most of the accuracy gap** (60 pp of the 73 pp total); the Corpus Adapter’s incremental accuracy gain over RAG is small on this sample and cannot be certified at $p < .05$. The Corpus Adapter’s remaining advantage is instead in *deployment properties* rather than raw accuracy: the vocabulary registry is a static YAML artefact (no embedding-model dependency at inference time, no top- K tuning, byte-level determinism across runs), it is human-auditable, and retargeting to a new domain is a config-file swap rather than a re-embedding pass. Per-item analysis of the six RAG misses attributes two to formatting drift (e.g. `providesPrivacyNotice` vs. `providePrivacyNotice`) that the Corpus Adapter’s post-hoc canonicaliser recovers, three to top- K retrieval misses that a full-registry constrained-selection prompt avoids by construction, and one to a genuine semantic near-miss that neither approach resolves.

6.5 Component Ablations

The pipeline stacks six sub-components on top of the four core stages: a heuristic *pre-filter* that admits deontic-marker candidates (§4), *prefilter hints* injected into

Table 3. Component ablations on the AIT corpus. Each row disables one pipeline component; all others are held fixed. LLM–regex agreement is a consistency proxy against a deterministic regex classifier (the paper’s original baseline of §6.1); it is not accuracy against gold because gold labels are baseline-specific and cannot be joined to ablation runs whose rule sets differ. Time is end-to-end wall-clock on the consumer profile host (§5.5).

Ablation	N rules	LLM–regex agr.	κ	Macro F1	Wall time
baseline	443	82.6%	.637	.748	34.9 min
no-prefilter	—	<i>timed out > 2h (evidence in itself)</i>			
no-hints	442	83.0%	.645	.750	95 min
no-reclassify	445	82.5%	.632	.745	47 min
no-fallback	445	82.5%	.632	.745	36 min
no-fol-retry	445	82.5%	.632	.745	36 min
no-may-disambig	499	78.4%	.592	.709	52 min

the Stage 2 classifier prompt, a Stage 2 *reclassify* second-opinion pass, a Stage 3 *FOL retry* loop on placeholder rejection, a Stage 4 *NL→SHACL fallback* for FOL-refused rules, and a “may” *disambiguation* filter that separates epistemic uncertainty from deontic permission. We disable each in turn and compare the resulting classified rule set against the baseline.

Table 3 separates the six components into three buckets. *Structurally necessary* components are the pre-filter, the “may” disambiguation filter, and the NL→SHACL fallback: the pre-filter is required for throughput (all 1,663 sentences reach the classifier without it, and a 2-hour budget expires before the run completes); disabling “may” disambiguation admits 56 additional candidates but LLM–regex agreement drops 4.2 pp, indicating that most of the extra admits are epistemic “may” constructions rather than genuine deontic permissions; and disabling the NL fallback drops 82 rules from the final shape set (18.5% coverage loss), because those rules never yield a well-formed FOL formula. *Empirically marginal* components on this metric are prefilter hints, the reclassify second-opinion pass, and the FOL retry loop: all three show swings smaller than the natural rule-count variation between runs (± 3 rules, agreement within ± 0.5 pp). We report this honestly rather than pruning the null results: the components may still be justified by orthogonal considerations (retry catches rare LLM parse failures that never surface at this metric, second-opinion addresses tail-risk classifications that only appear in adversarial inputs) but do not carry weight on classification consistency. The *no-prefilter timeout* is a design-defence result in its own right: the four-stage pipeline is not compute-tractable at corpus scale without the pre-filter, even on the near-8 GB consumer GPU that hosts the profile in §5.5. Full per-ablation JSON summaries are released in the artifact bundle (§8); the sweep script is `evaluation/full_ablation_sweep.py`.

6.6 Output Stability and External Validation

Stability (M5). Repeated pipeline runs with fixed seed and pinned model tags produce SHACL outputs hash-identical to the first run across all 443 shapes. This determinism is a prerequisite for deployments that need an audit trail tying a compliance check back to a specific policy interpretation.

LexDeMod external validation. On the LexDeMod [21] benchmark (D2: 200 lease-contract clauses) the same classifier drops from D1’s Macro $F_1 = .810$ to **Macro $F_1 = .370$** (obligation $F_1 = .569$; permission $F_1 = .038$). The failure has a diagnosable mechanism: lease contracts express permission as “*Tenant shall be entitled to ...*”, whose obligation-style “shall” cue dominates the LLM’s prompted heuristic, causing permissions to be absorbed into the obligation class. This is a vocabulary-prior mismatch, not an expressiveness or reasoning failure, and it is precisely the class of failure the Corpus Adapter is designed to address by registry swap. Building a lease-contract deontic-cue registry (analogous to the AIT-vs-GDPR comparison in §6.4) is the natural next experiment; the present LexDeMod number is the zero-shot lower bound against which any future adapter-on number must be compared. The boundary is consistent with our positioning: PolicyKG claims *institutional* sufficiency, not universal legal coverage, and a controlled cross-domain test under registry control is the GDPR experiment in §6.4.

7 Discussion

7.1 What Was Surprising

Three results ran against our prior expectations and deserve explicit comment.

The learned baseline beat the zero-shot LLM in-domain but not out-of-domain. A NumPy-only multinomial logistic regression on TF bag-of-words plus six deontic-marker features reaches Macro $F_1 = .863$ under 5-fold CV on D1, above the LLM’s zero-shot .810. We had assumed the reverse: a general-purpose 7B-parameter model exposed to trillions of tokens should beat a NumPy classifier trained on a few hundred sentences. On LexDeMod, however, the ordering flips into a tie: the same LR under 5-fold CV ($N = 200$) yields Macro $F_1 = .360$ against the LLM’s .370. The interpretation is that the LR was not learning “deontic logic”; it was learning the AIT-specific vocabulary distribution. When that distribution shifts to lease-contract English, the LR’s implicit prior evaporates and the LLM’s broader lexical coverage catches up. This is a warning against reading in-domain baseline superiority as evidence of generalisation; the opposite conclusion may be closer to the truth.

The permission-class collapse on LexDeMod ($F_1 = .038$) was larger than any effect we anticipated. The mechanism is diagnosable and mundane: lease contracts express permission through the phrase “Tenant shall be entitled to ...”, whose “shall” cue the LLM has learned to read as an obligation marker. Nearly every permission is absorbed into the obligation class. This is not a reasoning

failure and not an expressiveness failure; it is a vocabulary-prior mismatch, and it is precisely the failure mode the Corpus Adapter is designed to address at Stage 4. It also suggests, more broadly, that headline LLM performance on any deontic-classification benchmark should be reported against the lexical distribution of the target corpus, not against a single macro average.

The 0/443 SOL/HOL audit finding was cleaner than we expected. Going in, we assumed a small residue of quantifier-over-rule or predicate-over-predicate cases would surface in eligibility criteria or waiver clauses. None did. Even the six “exceptional” rules that looked meta-policy on first read, such as “the Registrar may waive the prerequisite in cases of documented hardship,” resolved to first-order predicates over an unquantified exception set. We report this as a lower-bound audit finding rather than a proof, but the null result is worth flagging: the institutional-policy sub-domain may be a less demanding target for formal-logic infrastructure than the theoretical literature has assumed.

7.2 Comparison to Prior Work

On the deontic-classification task, PolicyKG’s 86.9% D1 accuracy sits above the 75–85% range reported by pattern-based systems (Maxwell and Antón’s rule-model work at approximately 77% [16], Brodie et al.’s SPARCLE-based SVM at 82% on privacy policies [3]) and near PAPEL’s 80% F_1 on clause labelling with chain-of-thought prompting [8]. These are not head-to-head comparisons: each system targets a different corpus and a slightly different label set, so the accuracy delta is a directional indicator rather than a benchmark win. The substantive advance is not the accuracy number; it is the pipeline structure. Prior LLM-for-legal work (LexGLUE [4], LegalBench [11], PAPEL [8]) delivers labels or spans; PolicyKG delivers executable SHACL shapes with an audit trail from a sentence to its formula to its shape. The chain is what enables downstream compliance checking rather than downstream annotation.

On the FOL-sufficiency question, our contribution is a resolution level, not a direction. Governatori and Rotolo [10] argued for FOL adequacy on theoretical grounds; Brachman and Levesque [2] framed the expressiveness-tractability trade-off. Neither line had been tested against a real institutional corpus at scale. Our 0/443 checklist audit provides that empirical anchor. It confirms rather than contradicts the theoretical argument, but it turns an “it should suffice” into “on this corpus it did suffice.” That is a smaller but different type of claim.

On vocabulary grounding, the Corpus Adapter sits alongside constrained-decoding work (Willard and Louf’s Outlines [23], Geng et al.’s grammar-constrained decoding [7]) but operates at a different abstraction level. Those approaches constrain *syntax* at the token level; the Corpus Adapter constrains *vocabulary* at the predicate level, via a runtime-editable registry rather than a compiled grammar. The two are compositional and the extended journal version will report a stacked ablation.

7.3 Experimental Decisions Worth Revisiting

Three experimental decisions traded ambition for tractability, and a reviewer would be right to press on each.

Zero-shot only for the main LLM. We ran the classification and formalisation stages zero-shot to isolate the pipeline’s architectural contribution from prompt-engineering gains. Few-shot with representative in-context exemplars would plausibly close much of the gap to the LR baseline on D1, but it would also introduce a selection-bias question (which exemplars, drawn from what distribution?) that we chose not to entangle with the primary claim. The extended version will report a matched few-shot condition.

$N = 15$ for the GDPR transfer experiment. The N was sized to isolate the registry-swap effect on a hand-curated, ontology-anchored rule set, not to certify GDPR coverage. Under a denominator of 1 in the no-adapter condition, the fold-change ratio is undefined and the Wilson CI is wide; we accordingly report absolute proportions, risk difference, and Cohen’s $h = 1.53$ rather than a ratio. The Fisher’s exact $p < .001$ remains valid at this N and the effect size is large enough that a bigger sample is unlikely to reverse the sign, but a 100+ rule set drawn from the full GDPR would strengthen the external-validity argument.

Single LLM in the main results. Mistral 7B Instruct was selected from a five-model pilot for zero parse-error rate and inference stability, not for the highest raw accuracy. Reviewers will reasonably ask whether the Corpus Adapter’s cross-domain gain is a property of Mistral or a property of the pipeline. The pilot data on the 50-item IRR sample is consistent across five instruction-tuned models (Fleiss’ $\kappa = .635$ across LLMs plus human gold), but a matched adapter on/off contrast on Gemma 3, Qwen 3, and a 70B-parameter model is on the roadmap and will settle the question directly.

7.4 Reconciling the In-Domain and Cross-Domain Stories

Bringing the strands together: when the target domain comes with sufficient labelled data, a learned baseline is the right tool; when it does not (the standard deployment scenario for institutional policy formalisation), the LLM-plus-Corpus-Adapter combination is the only option. An LLM/LR ensemble on D1 is appealing but undefined off-domain (no training signal), and the Corpus Adapter is not substitutable into a learned baseline because it constrains predicate *generation* (Stage 3–4), not category *classification* (Stage 2). The two roles are orthogonal, and the deployment case is the case where only one of the two is available.

7.5 Deployment: Compliance Dashboard

To demonstrate that the pipeline output is directly consumable by a compliance workflow rather than only by downstream research, we release a companion web dashboard (FastAPI backend, HTML/JavaScript frontend; ~33 KB, in `web/app.py`) with the artifact bundle. The dashboard loads the pipeline’s

SHACL shapes and lets a compliance officer (i) browse the 443 classified rules with filters by deontic type and severity, (ii) inspect each rule’s source PDF span, generated FOL formula, and emitted SHACL shape side-by-side, (iii) select individual entities (students, faculty, staff, committees) from a seeded PostgreSQL demonstration database via `db/rdf_converter.py`, and (iv) run `pyshacl` validation in-browser to surface entity-centric violations with severity coding and property-level explanations. The 42 syntactically invalid shapes from the NL-fallback path are automatically skipped at load time. The dashboard is a compliance-workflow demonstration, not an evaluation artifact; its role in the paper is to show that the audit-trail requirement (§1) is instantiated end-to-end from policy PDF to entity-level violation report.

7.6 Limitations

The FOL-sufficiency finding (0 of 443 SOL/HOL matches) is a *checklist-based audit*, a lower bound on FOL adequacy scoped to our institutional sub-domain and to FOL with standard arithmetic/temporal extensions. Beyond that scope, the evaluation is limited along five axes: *single institution* (one university; $N = 15$ GDPR transfer partly mitigates); *English only*; *declarative rules only* (no procedural workflows or rule-set meta-reasoning); *static validation* (SHACL presupposes an upstream enforcement layer); *reference-set construction* (in-house, so M4 is best read as the $F_1 = .000 \rightarrow .866$ adapter delta, not as performance against an independent gold). LexDeMod permission $F_1 = .038$ bounds the zero-shot transfer limit for the permission class and motivates a lease-contract registry as future adapter demonstration. A representative NL-fallback failure: the rule “Continuing students may put their names on the waiting list *and* shall be served on a queue-numbering basis *or*, exceptionally, through email to OFAM” compounds two deontic operators with a nested disjunction that exceeds single-pass Stage 3 parsing.

8 Conclusion and Future Work

We presented PolicyKG, an agentic LLM pipeline that translates institutional policy documents into SHACL-validated knowledge graphs via a first-order deontic intermediate layer. The pipeline achieves 86.9% deontic classification accuracy (Fleiss’ $\kappa = .844$ inter-annotator, 95% CI [.67, .97]), $F_1 = .866$ SHACL shape correctness on a 69-shape evaluation subset, and a two-pass audit (regex checklist over all 443 rules plus manual review of the 92 FOL-fallback cases) finding no SOL/HOL constructs across the AIT corpus (exact upper 95% CI on the true HOL rate: 0.67%). The main technical contribution, the Corpus Adapter, enables cross-domain transfer: in a controlled GDPR experiment ($N = 15$), exact vocabulary alignment rises from 1/15 to 11/15 (Fisher’s exact $p < .001$, Cohen’s $h = 1.53$).

The broader insight is that for LLM-based knowledge-graph construction, *vocabulary grounding via configuration beats free generation*. Reframing predicate

emission as constrained selection from a registry is cheaper than fine-tuning, transfers immediately across domains, and leaves an audit trail. The pattern applies wherever LLMs must emit canonical identifiers from a closed-world ontology, from biomedical knowledge graphs to legal taxonomies to `schema.org`-grounded extraction. A learned in-domain baseline can exceed the zero-shot LLM on the source corpus, but only the LLM-plus-adapter combination generalises to a new corpus without retraining; the extended journal version will report a real adapter on/off LLM ablation, a LegalBERT fine-tuned baseline, and a scaled GDPR portability set.

Future work. Four directions extend this work.

Multi-institutional validation. The 0/443 SOL/HOL audit is scoped to one university’s policy corpus. A meaningful test of the broader FOL-sufficiency claim requires replicating the audit across policy corpora from at least three additional institutions that differ in size, jurisdiction, and academic type (public research university, private undergraduate college, professional-school setting). We expect the null result to hold; a single institution that produces even one SOL/HOL construct would sharpen the architectural claim about where FOL falls short.

Defeasible deontic extensions. A subset of institutional rules carry implicit exceptions (“the Registrar may waive the prerequisite. . .”) that our current FOL grammar treats as first-order predicates over an unquantified exception set. A defeasible deontic layer, following the tradition of Governatori and colleagues [9], would represent the override relation explicitly, enabling SHACL shapes to encode “normally prohibited, but with documented cause” semantics rather than flatten the exception into a boolean predicate. This is a natural extension because Stage 3’s structured JSON schema already carries an `exception_of` field that is currently unused downstream.

Agentic SHACL repair. On validation failure, the pipeline currently retries once with the validator’s diagnostic. A stronger loop would invoke the pipeline in reverse: an LLM reads the SHACL report, walks back through the FOL formula to the source sentence, and proposes either a data-side remediation (which entity attribute is missing) or a shape-side one (which constraint is over-strict). The auditability of the FOL intermediate is the mechanism that makes this practical: the LLM has a symbolic anchor to reason over rather than a free-form policy sentence.

Scaled and stacked ablations. The extended journal version will report a matched adapter on/off LLM ablation, a LegalBERT fine-tuned baseline, a few-shot LLM condition, a 100+-rule GDPR transfer set, and a stacked comparison against grammar-constrained decoding to disentangle syntax-level from vocabulary-level grounding. These are the experiments a reviewer would ask for; we would rather run them than defend against the ask.

Code and Data Availability. All artifacts referenced in this paper (pipeline source, AIT and GDPR Corpus Adapter registries, post-audit gold annotations, the 12-correction script, logic-expressiveness audit, Fleiss- κ bootstrap, multi-tier

shape-correctness evaluation, Corpus Adapter ablations, multi-LLM pilot data, offline learned baseline, and the frozen output snapshot) are released as a single bundle at Zenodo (DOI placeholder pending camera-ready) with a GitHub mirror.

Acknowledgments. This work was supported by a Royal Thai Government Fellowship. The authors thank the AIT Brain Lab for compute and discussion, and Kittipat (Law / Legal Studies background) and Mayuree (Linguistics background) for serving as the two external annotators in the inter-rater reliability study.

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